

HD189733b

R-0057

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HD189733_230813 · created 2026-07-15T13:26:59+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460169.751 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.187 +/- 0.054
Transit depth (Rp/Rs)^2	3.5 +/- 2.01 [%]
Orbital Inclination (inc)	85.64 +/- 2.64
Airmass coefficient 1 (a1)	0.676 +/- 0.069
Airmass coefficient 2 (a2)	0.35 +/- 0.61
Scatter in the residuals of the lightcurve fit is	9.88 %
Best Comparison Star	#1 - [209, 122]
Optimal Aperture	3.85
Optimal Annulus	11.36
Transit Duration (day)	0.068 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.187 ± 0.054 vs 0.15534 (deviation 0.56σ)
Duration fit vs published	0.068 d vs 0.0754 d (deviation 0.27σ)
Mid-time O–C	-4.2 min
Depth SNR	3.3
Residual scatter	9.88 %

Light curve

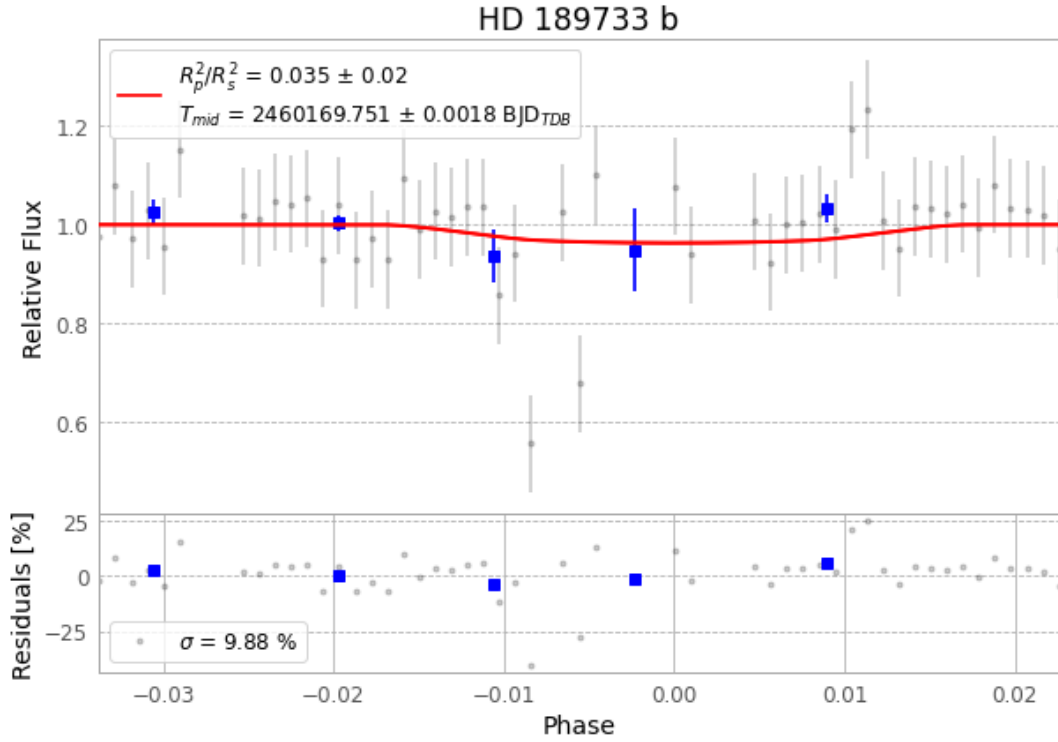


Figure 1 — FinalLightCurve_HD 189733 b_2023-08-12.png (SHA-256 431e5870db0f393d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [246, 217], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 50b1a7cfa5e8a7d0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

60 FITS frames (39 MB), HD189733230813040615.FITS → HD189733230813070615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HD189733-230813.log (99689dd24930...), FinalLightCurve_HD 189733 b_2023-08-12.png (431e5870db0f...), FinalLightCurve_HD 189733 b_2023-08-12.csv (fb7537f43bf7...)

Compute resources

Wall time 254.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.